Conformal Prediction

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1 Introduction

Conformal prediction is a relatively new framework for quantifying uncertainty in the predictions made by arbitrary prediction algorithms. Fundamentally, it does so by converting an algorithm's predictions into prediction sets, which have strong finite-sample coverage properties.

The idea behind conformal prediction was born out of conversations between Vladimir Vovk and his colleagues, including Alexander Gammerman and Vladimir Vapnik (who was visiting the university), in the mid 1990s at Royal Holloway, University of College London. The definitive reference is Vovk et al. (2005). This remained a topic of intense interest for Vovk and collaborators for many years, up through current day. It is Larry Wasserman who brought about the interest in the statistics community, and inspired Carnegie Mellon University colleagues including Jing Lei (and the author of these lecture notes), to collaborate on conformal prediction in the mid 2010s. This lecture will draw mostly from the language for conformal prediction developed in Lei et al. (2018); Tibshirani et al. (2019).

The community working on conformal prediction remained fairly small for most of its short history, until fairly recently—just the last few years, really—when it exploded in popularity in the machine learning community. As such, there will be a lot of interesting work about conformal prediction that we will not be covering. A nice recent overview is Angelopoulos and Bates (2023). The last section of that monograph provides some sense of the current trends in the field.

1.1 A lofty goal?

The basic goal of conformal prediction is as follows. Let $(X_i, Y_i) \sim P$, $i = 1, \ldots, n$ be i.i.d. feature and response pairs, from a distribution P on $\mathcal{X} \times \mathcal{Y}$. For concreteness, we can think of the feature space as say, $\mathcal{X} = \mathbb{R}^d$, and the response space as $\mathcal{Y} = \mathbb{R}$, though this need not be the case in general. Let $\alpha \in (0, 1)$ be and a nominal error level. Then we want to find a *prediction band*,

$$\hat{C}_n : \mathcal{X} \to \{\text{subsets of } \mathcal{Y}\},$$

with the property that for a new i.i.d. pair $(X_{n+1}, Y_{n+1}) \sim P$,

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \geq 1 - \alpha, \tag{1}$$

where the probability is over all of our data (X_i, Y_i), $i = 1, \ldots, n + 1$.

On the one hand, without placing any assumptions on P, and without appealing to asymptotics of any kind, this might seem like a really hard goal in general. On the other hand, we can do something trivial to obtain it: for example,

$$\hat{C}_n(X_{n+1}) = \begin{cases} \mathcal{Y} & \text{with probability } 1 - \alpha \\ \emptyset & \text{with probability } \alpha \end{cases}$$

will always have exactly $1 - \alpha$ coverage, that is, it will achieve (1) with an equality.

So the real question is this (albeit still somewhat vaguely-phrased): can we achieve (1), in finite samples, without any assumptions on P, by doing something "nontrivial"? In particular, we would like our strategy to adapt to the hardness of the problem, in the following sense: the more easily we can predict Y_{n+1} from X_{n+1}, the smaller we would like our set $\hat{C}_n(X_{n+1})$ to be.

1.2 This is achieveable!

Perhaps remarkably, this last goal is actually achieveable, in a very general way. As we will see in the coming sections, we can start with any algorithm that produces a "point predictor" $\hat{f}_n$ that predicts Y_{n+1} from X_{n+1}, and turn this into a "set predictor" $\hat{C}_n$ that satisfies (1).

The basic idea behind conformal prediction is two-fold. The first key idea can actually be explained in a simpler context, *where there are no features at all*, and we just have a sequence $Y_i \in \mathbb{R}$, $i = 1, \ldots, n$ of real-valued response values. Suppose our goal is to find a one-sided prediction interval $\hat{C}_n = (-\infty, \hat{q}_n]$ with

$$\mathbb{P}\big(Y_{n+1} \le \hat{q}_n\big) \ge 1 - \alpha. \tag{2}$$

Given this goal (2), a natural place to start would be to set $\hat{q}_n$ to be the level $(1-\alpha)$ sample quantile of $Y_1, \ldots, Y_n$, which we denote by

$$\hat{q}_n = \text{Quantile}\bigg(1 - \alpha;\ \frac{1}{n}\sum_{i=1}^n \delta_{Y_i}\bigg),$$

with δ_a denoting a point mass at a, and hence $\frac{1}{n}\sum_{i=1}^n \delta_{Y_i}$ denoting the empirical distribution of $Y_1, \ldots, Y_n$. But this would only give use the approximate result

$$\mathbb{P}(Y_{n+1} \le \hat{q}_n) \approx 1 - \alpha.$$

This becomes exact as $n \to \infty$, under standard conditions (that ensure convergence of the sample quantile to the population quantile). So can we instead get something that satisfies (2) in finite-sample?

First key idea: use ranks to form adjusted quantiles. This is where the first key idea behind conformal prediction comes in (which in a sense traces back to work on rank-based statistics and permutations by Fisher and Pitman in the 1930s). As Y_{n+1} is i.i.d. with $Y_1, \ldots, Y_n$, then

$$\text{the rank of } Y_{n+1} \text{ is uniformly distributed over the values } Y_1, \ldots, Y_{n+1}. \tag{3}$$

This means that

$$\mathbb{P}\Big(Y_{n+1} \text{ is among the } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_{n+1}\Big) \ge 1 - \alpha, \tag{4}$$

which is in turn equivalent to[1]

$$\mathbb{P}\Big(Y_{n+1} \text{ is among the } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_n\Big) \ge 1 - \alpha. \tag{5}$$

The last step is critical: note that we have moved from a comparison between Y_{n+1} and a an order statistic of $Y_1, \ldots, Y_{n+1}$ in (4) to a comparison between Y_{n+1} and an order statistic of $Y_1, \ldots, Y_n$ in (5). This is key, because what is on the right-hand side of the $\le$ sign in (5) is *computable from just the first n points.* Accordingly, by defining

$$\hat{q}_n = \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_n, \tag{6}$$

we have precisely achieved (2).

The formulation in (6) is arguably the most intuitive way to remember how to achieve coverage. There are other equivalent formulations. One such equivalent formulation (we will see more later on) is

$$\hat{q}_n = \text{Quantile}\bigg(\frac{\lceil (1-\alpha)(n+1) \rceil}{n};\ \frac{1}{n}\sum_{i=1}^n \delta_{Y_i}\bigg). \tag{7}$$

[1] To see this, consider the complement of the events (inside the probabilities) in (4), (5). Abbreviate $k = \lceil (1-\alpha)(n+1) \rceil$. Then $Y_{n+1} >$ the k smallest of $Y_1, \ldots, Y_{n+1}$ is clearly an equivalent statement to $Y_{n+1} >$ the k smallest of $Y_1, \ldots, Y_n$, since Y_{n+1} can never be strictly larger than itself. That said, this argument really only makes sense for $k \le n$, and for $k = n+1$, which occurs if $\alpha < 1/(n+1)$, then $\lceil (1-\alpha)(n+1) \rceil = n+1$, then we have to interpret the $(n+1)$ smallest of $Y_1, \ldots, Y_n$ as being $+\infty$ to equate (4), (5). This is the consistent with interpreting the quantile function in (7) to return $+\infty$ when the input level is ≥ 1.

In this way, we can also see the solution here as simply to taking the sample quantile at an adjusted level: we use $\lceil (1-\alpha)(n+1) \rceil / n$, instead of $1-\alpha$, which is a sort of finite-sample correction. But in our opinion, the fact that (7) achieves the coverage guarantee is less obvious; only through its equivalence to (6)—and then the equivalence to the precedings displays in (5), (4), (3)—does this become transparent. A very simple illustration of the key idea here is given in Figure 1.

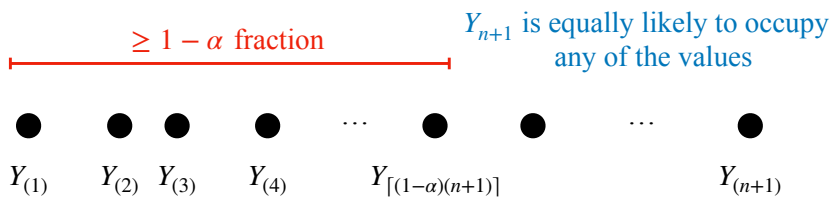

Figure 1: *Illustration of the first key idea in conformal prediction, as stated in* (3), (4). *Note also that we have the sharpened version* (8) *when there are almost surely no ties.*

~~Love~~ Exchangeability is all you need. Looking back at (3), all that we need for this to hold is that $Y_1, \ldots, Y_{n+1}$ are *exchangeable*, which is a weaker than the i.i.d. assumption. Recall that exchangeability of $Y_1, \ldots, Y_{n+1}$ means that their joint distribution is unchanged under permutations:

$$(Y_1, \ldots, Y_{n+1}) \stackrel{d}{=} (Y_{\sigma(1)}, \ldots, Y_{\sigma(n+1)}), \quad \text{for all permutations } \sigma.$$

Coverage upper bound when there are no ties. If there are almost surely no ties between $Y_1, \ldots, Y_{n+1}$ (or we use a suitably random tie-breaking rule) then the statement in (4) can be sharpened to an equality,

$$\mathbb{P}\Big(Y_{n+1} \text{ is among the } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_{n+1}\Big) = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1}. \tag{8}$$

Simply upper bounding the right-hand side above gives

$$\mathbb{P}\Big(Y_{n+1} \text{ is among the } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } Y_1, \ldots, Y_{n+1}\Big) < (1-\alpha) + \frac{1}{n+1}. \tag{9}$$

Carrying on from by the same logic as before leads to the sharpened conclusion,

$$\mathbb{P}\big(Y_{n+1} \leq \hat{q}_n\big) \in \Big[1-\alpha,\, 1-\alpha+\frac{1}{n+1}\Big). \tag{10}$$

with $\hat{q}_n$ still defined as in (6). To be clear, the lower bound on coverage in (10) always holds, and the upper bound holds under the assumption that there are almost surely no ties.

Naive attempt to lift this idea to regression problems. Now let's try to lift the first key idea to a regression setting, where we observe both $X_i \in \mathcal{X}$ and $Y_i \in \mathbb{R}$, $i = 1, \ldots, n$, and want a prediction set for Y_{n+1} based on X_{n+1}. Suppose that $\hat{f}_n$ is any point predictor, trained on (X_i, Y_i), $i = 1, \ldots, n$, such that (to put it informally)

$$\hat{f}_n(x) \text{ predicts the value of } y \text{ that we expect to see at } x.$$

Then we could proceed naively as follows. We define (absolute) residuals made on the training set,

$$R_i = |Y_i - \hat{f}_n(X_i)|, \quad i = 1, \ldots, n,$$

and just as in (6), let $\hat{q}_n = \lceil (1-\alpha)(n+1) \rceil$ smallest of $R_1, \ldots, R_n$. We could then define the prediction set to be $\hat{C}_n(x) = \{y : |y - \hat{f}_n(x)| \leq \hat{q}_n\}$, or in other words

$$\hat{C}_n(x) = \big[\hat{f}_n(x) - \hat{q}_n,\, \hat{f}_n(x) + \hat{q}_n\big],$$

in the hope that $Y_{n+1} \in \hat{C}_n(X_{n+1})$ with probability at least $1-\alpha$. However,

$$Y_{n+1} \in \hat{C}_n(X_{n+1}) \iff R_{n+1} \le \hat{q}_n \iff R_{n+1} \le \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } R_1, \dots, R_n,$$

and the latter event does *not* hold with probability $1-\alpha$, because $R_{n+1} = |Y_{n+1} - \hat{f}_n(X_{n+1})|$ is *not* exchangeable with $R_1, \dots, R_n$.

The problem is that $\hat{f}_n$ has already "seen" (X_i, Y_i), $i = 1, \dots, n$ (since it was trained on them), but it has not yet seen (X_{n+1}, Y_{n+1}). Accordingly, the test residual R_{n+1} will be generally stochastically larger than the training residuals $R_1, \dots, R_n$, and so the naive prediction set defined above will generally undercover.

2 Split conformal prediction

Enter the second key idea behind conformal prediction, and *split conformal prediction*, which is the simplest and most computationally efficient way to carry out this idea. Split conformal prediction is the focus of this section, and we stick to regression where responses are real-valued, so $\mathcal{Y} = \mathbb{R}$. The next section describes a (much more complicated) method that avoids splitting the data. At the end of this lecture, we will consider classification, where $\mathcal{Y}$ is discrete.

Second key idea: construct scores symmetrically. In a nutshell, the second key idea in conformal prediction is to build residuals in way that treats all of the data (that goes into determining their distribution), including the test data, in a *symmetric* fashion. This will ensure that the residuals obey the exchangeability condition we require in order to get coverage.

Concretely, in split conformal prediction (split CP) we do the following. We first divide the training set into two sets:

- D_1, called the *proper training set*; and
- D_2, called the *calibration set*.

We can think of these as sets of indices, so that D_1, D_2 are such that $D_1 \cap D_2 = \emptyset$ and $D_1 \cup D_2 = \{1, \dots, n\}$. Let $n_1 = |D_1|$ and $n_2 = |D_2|$. The next step is to fit our point predictor on proper training points (X_i, Y_i), $i \in D_1$, call it $\hat{f}_{n_1}$. Then we define calibration set residuals

$$R_i = |Y_i - \hat{f}_{n_1}(X_i)|, \quad i \in D_2,$$

a conformal quantile

$$\hat{q}_{n_2} = \lceil (1-\alpha)(n_2+1) \rceil \text{ smallest of } R_i,\ i \in D_2,$$

and a conformal set

$$\hat{C}_n(x) = \big[\hat{f}_{n_1}(x) - \hat{q}_{n_2},\ \hat{f}_{n_1}(x) + \hat{q}_{n_2}\big], \tag{11}$$

The main guarantee we can get is that

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(X_{n+1}) \,\Big|\, (X_i, Y_i),\ i \in D_1\Big) \in \Big[1-\alpha,\ 1-\alpha+\frac{1}{n_2+1}\Big), \tag{12}$$

where the lower bound on coverage always holds, and the upper bound holds under the assumption that the residuals are almost surely distinct. Why? If we condition on the proper training set (X_i, Y_i), $i \in D_1$, then *the calibration residuals R_i, $i \in D_2$ and the test residual $R_{n+1} = |Y_{n+1} - \hat{f}_{n_1}(X_{n+1})|$ are all i.i.d.*, and therefore

$$Y_{n+1} \in \hat{C}_n(X_{n+1}) \iff R_{n+1} \le \hat{q}_{n_2} \iff R_{n+1} \le \lceil (1-\alpha)(n_2+1) \rceil \text{ smallest of } R_i,\ i \in D_2$$

occurs (by our previous arguments) with probability at least $1-\alpha$, and at most $1-\alpha+1/(n_2+1)$ if the residuals are almost surely distinct.

Any score function will work. Above, we used absolute residuals as a negatively-oriented score function (negatively-oriented meaning that lower values are better). But really, any negatively-oriented score function will do, and the argument goes through just as before. That is, suppose $V(x,y) = V((x,y); \hat{f}_{n_1})$ assigns a conformity score to the point (x,y) based on $\hat{f}_{n_1}$ (for simplicity, we will generally drop the notational dependence on $\hat{f}_{n_1}$). Define, generalizing the construction leading up to (11), calibration set scores

$$R_i = V(X_i, Y_i), \quad i \in D_2,$$

and a conformal set

$$\hat{C}_n(x) = \Big\{ y : V(x,y) \le \lceil (1-\alpha)(n_2+1) \rceil \text{ smallest of } R_i,\, i \in D_2 \Big\}.$$

Then we get the exact same guarantee as in (12), since R_i, $i \in D_2$ and $R_{n+1} = V(X_{n+1}, Y_{n+1})$ are all still i.i.d., conditional on $\hat{f}_{n_1}$). This will be important later, as we'll see how to move beyond the residual score to obtain better local adaptivity in our prediction bands.

Positively-oriented scores will work too, we can just negate them (to make the negatively-oriented) before passing them through this construction. This would result in a conformal set of the form

$$\hat{C}_n(x) = \Big\{ y : V(x,y) \ge \lfloor \alpha(n_2+1) \rfloor \text{ smallest of } R_i,\, i \in D_2 \Big\}.$$

An example of a naturally occurring positively-oriented scores will arise in the classification setting.

Quantile and CDF formulations. Keeping the conformity score generic (and negatively-oriented) for now, we note that there are multiple equivalent formulations for the split conformal prediction set:

$$\hat{C}_n(x) = \Big\{ y : V(x,y) \le \lceil (1-\alpha)(n_2+1) \rceil \text{ smallest of } R_i,\, i \in D_2 \Big\} \tag{13}$$

$$= \left\{ y : V(x,y) \le \text{Quantile}\Big(\frac{\lceil (1-\alpha)(n_2+1) \rceil}{n_2}; \frac{1}{n_2} \sum_{i \in D_2} \delta_{R_i} \Big) \right\} \tag{14}$$

$$= \left\{ y : \frac{1}{n_2} \sum_{i \in D_2} 1\{R_i \le V(x,y)\} \le \frac{\lceil (1-\alpha)(n_2+1) \rceil}{n_2} \right\}. \tag{15}$$

The original formulation in (13) is (we think) the most intuitive.

The second formulation in (14) is of the form "test score $\le$ adjusted quantile". This form will be useful for generalizing conformal prediction to use weights, which we will cover in the next lecture, when we consider certain settings where the scores are no longer exchangeable.

The third formulation (15) is expressed in terms of the empirical cumulative distribution function (CDF) of R_i, $i \in D_2$. It is of the form "CDF evaluated at test score $\le$ adjusted level". This is a useful form when considering auxiliary randomization schemes, covered next.

Auxiliary randomization to get exact coverage.* It is worth noting that we can always use auxiliary randomization to get exact coverage in our prediction sets, that is, to achieve $1-\alpha$ coverage in (12). You can skip this without interrupting the flow of understanding the ideas in the rest of this lecture, hence the asterisk. First, rewrite the conformal set in its CDF form (15) as

$$\hat{C}_n(x) = \left\{ y : \frac{1}{n_2+1} \sum_{i \in D_2} 1\{R_i \le V(x,y)\} + \frac{1}{n_2+1} \le \frac{\lceil (1-\alpha)(n_2+1) \rceil}{n_2+1} \right\}.$$

This now compares the empirical CDF on the n_2+1 points: R_i, $i \in D_2$ and $V(x,y)$, to an adjusted level. To explain the auxiliary randomization mechanism, it helps to look at what happens at the (unknown) test point (X_{n+1}, Y_{n+1}): let $R_{n+1} = V(X_{n+1}, Y_{n+1})$ denote its score, and let $\hat{F}_{n_2+1}$ denote the CDF of R_i, $i \in D_2$ and R_{n+1}. Then

$$Y_{n+1} \in \hat{C}_n(X_{n+1}) \iff \hat{F}_{n_2+1}(R_{n+1}) \le \frac{\lceil (1-\alpha)(n_2+1) \rceil}{n_2+1},$$

which we know occurs with probability at least $1-\alpha$.

In general, if Z has CDF F, then $F(Z)$ is always sub-uniform, meaning $\mathbb{P}(F(Z) \leq t) \leq t$ for any $t \in [0,1]$. This is another way of seeing the coverage result stated above. Meanwhile, if F is continuous, then $F(Z)$ will be precisely uniform, $\mathbb{P}(F(Z) \leq t) \leq t$ for any $t \in [0,1]$. Furthermore, we can always take any CDF and make it continuous by injecting auxiliary randomness, defining for $U \sim \text{Unif}(0,1)$,

$$F^*(z) = \lim_{y \to z^-} F(y) + U \cdot \Big(F(z) - \lim_{y \to z^-} F(y) \Big).$$

Returning to our conformal setting, we see that if we define a randomized conformal prediction set based on auxiliary randomization of the empirical CDF, such that

$$Y_{n+1} \in \hat{C}_n^*(X_{n+1}) \iff \hat{F}_{n_2+1}^*(R_{n+1}) \leq 1-\alpha,$$

then we will get coverage with probability exactly $1-\alpha$. Spelling it out in more detail, this is the same as defining the randomized conformal set

$$\hat{C}_n^*(x) = \bigg\{ y : \frac{1}{n_2+1} \sum_{i \in D_2} 1\{R_i < V(x,y)\} + \frac{U}{n_2+1} \bigg(\sum_{i \in D_2} 1\{R_i = V(x,y)\} + 1 \bigg) \leq 1-\alpha \bigg\},$$

where $U \sim \text{Unif}(0,1)$ is independent of everything else. To record its guarantee, this set always satisfies

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n^*(X_{n+1}) \,\Big|\, (X_i, Y_i),\, i \in D_1 \Big) = 1-\alpha.$$

2.1 Remarks

Here we make some brief remarks about split conformal prediction. First, recall that the naive prediction band—as covered at the end of Section 1.2—is generally going to undercover, drastically so when $\hat{f}_n$ overfits the training data. In a sense, we can think of the split conformal band (11) as being *protected against overfitting*, since it based on comparing a test score to calibration set scores, and in the overfitting regime, these will all be equally large (in distribution).

Second, note that the split conformal band (11) constructed using absolute residual scores has width that is *exactly constant in* x. This is not generally a good thing: it means that the band does not adapt to the local hardness of the problem (how hard it is to predict at any given x), as we will clearly see in an example to follow. However, this can be addressed by changing the conformity score, as we will do in Section 4.

Third and last, we note the following key fact: *the better the point predictor* $\hat{f}_{n_1}$ (from the proper training set), *the tighter the prediction band will be*. Both experiments and theory corroborate this claim; see, e.g., Lei et al. (2018), and Figure 2, which is taken from that paper. (Do not confuse this point with the last point about local width at a particular x; here we are talking about the of the prediction band width in an *average* sense over x.) Therefore, any prediction algorithm leads to valid coverage, but better algorithms (for the prediction problem at hand) lead to smaller prediction sets.

An interesting way to interpret this last observation is as follows. Let us condition on the proper training set implicitly so we do not have to express it notationally. Then, average length is:

$$\mathbb{E}_{(X_i,Y_i) \sim P,\, i \in D_2} \bigg[\int \int_{\hat{C}_n(x)} d\mu(y)\, dP_X(x) \bigg]$$

where μ is Lebesgue measure and P_X is the distribution of X, the feature vector. Meanwhile, coverage is:

$$\mathbb{E}_{(X_i,Y_i) \sim P,\, i \in D_2} \bigg[\int \int_{\hat{C}_n(x)} dP_{Y|X}(y)\, dP_X(x) \bigg],$$

where $P_{Y|X}$ is the distribution of $Y|X$. Therefore, an inefficient prediction algorithm must somehow put mas in *low density regions* of $P_{Y|X}$, which does not hurt its coverage, but inflates its length.

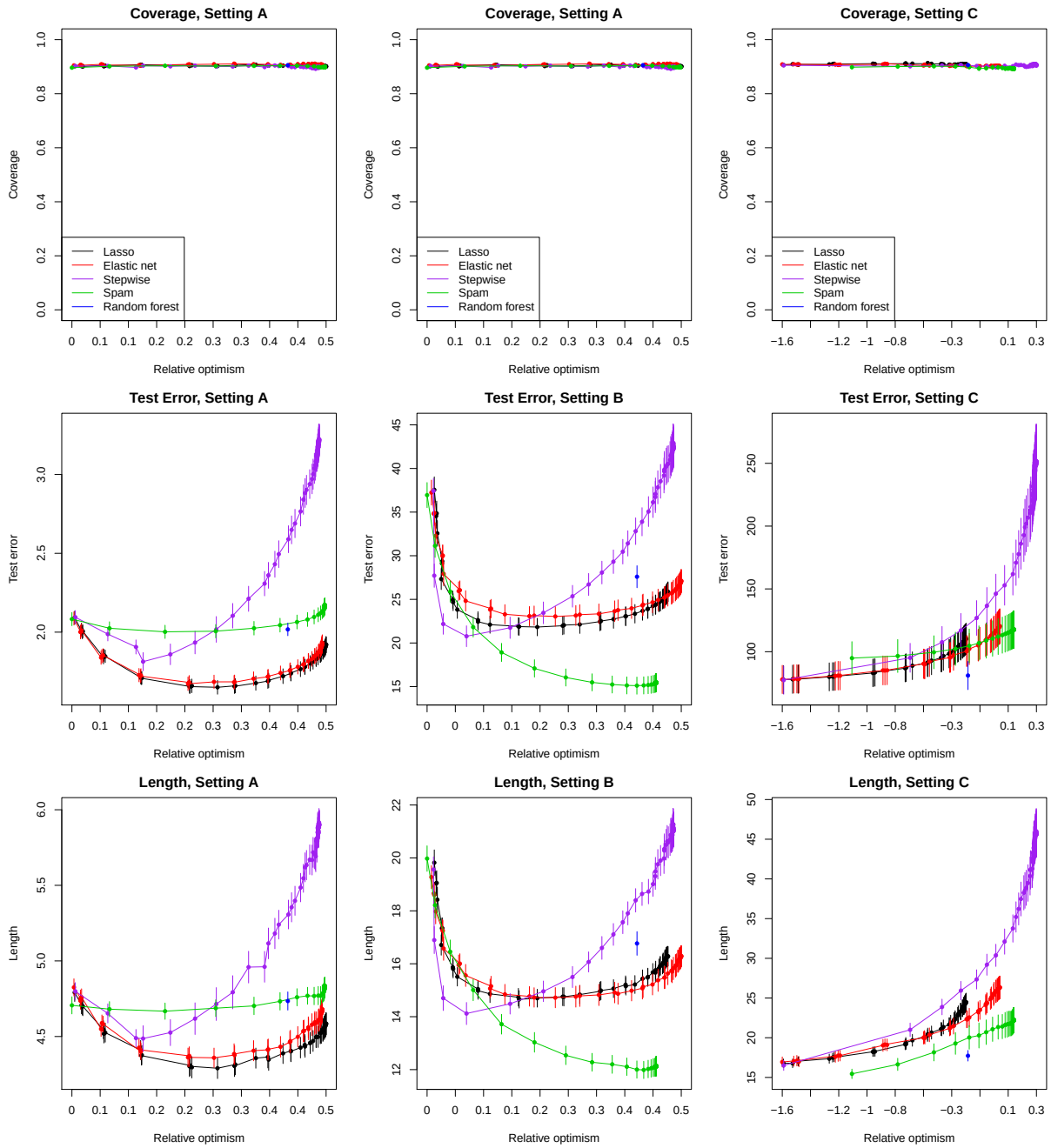

Figure 2: *Experiments demonstrating the coverage (top row), test error (middle row), and average length (bottom row) of split conformal prediction in three different simulation settings, and with several different prediction algorithms. The x-axis in each plot sweeps over internal hyperparameters of the algorithms (they are simply put on a common scale using a notion called relative optimism). Settings A, B, and C are increasingly challenging, in terms of the tractability of prediction. Takeaways: any algorithm, using any hyperparameter value, leads to essentially exactly 90% coverage in all settings (this was the nominal level), as seen in the top row; moreover, test error of the algorithm-hyperparameter pair and average length (or width, these are used synonymously) of the prediction set correlate quite strongly. Credit: Lei et al. (2018).*

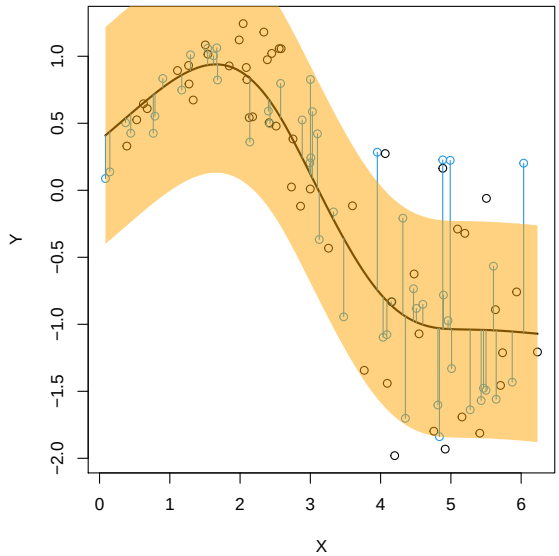

Figure 3: *Example of split conformal prediction, based on a smoothing spline with 5 degrees of freedom.*

2.2 Example

Now we give an example of split conformal in action, in Figure 3. In this simple univariate example (real-valued features), we split the data randomly and equally into a proper training set (points drawn in black) and a calibration set (drawn in blue). We use a smoothing spline with 5 degrees of freedom to fit $\hat{f}_{n_1}$ on the proper training set. The split conformal prediction band, which is simply computed from an adjusted level 90% quantile of the calibration residuals, and is drawn in orange.

This is guaranteed to have at least 90% test coverage, when we draw test points from the same distribution as that used to generate the training data. We can see that the band is constant-width, by design. This is a function of using the absolute residual as our conformity score. It is not desirable in the current example because it will tend to overcover on the left side of the domain, and undercover on the right side (note that the variance of $Y|X$ is not constant in this example). We will revisit (and remedy) this later in Section 4.

2.3 Conditional coverage properties?

We have seen that split conformal prediction comes with the strong, distribution-free coverage guarantee in (12). Of course, simply by marginalizing over the proper training set, it also has the unconditional coverage property,

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(X_{n+1})\Big) \in \Big[1-\alpha,\, 1-\alpha+\frac{1}{n_2+1}\Big).$$

Going the other direction, we could ask if it has coverage properties after we condition on *more* than just the proper training set. If we condition on both the proper training set and the calibration set, that is, we condition on the *entire training set*, then when the conformity scores are almost surely distinct (or we use a suitably random tie-breaking rule):

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(X_{n+1}) \,\Big|\, (X_i, Y_i),\, i = 1, \ldots, n\Big) \sim \text{Beta}(k_\alpha, n_2 + 1 - k_\alpha), \tag{16}$$

where $k_\alpha = \lceil (1-\alpha)(n_2+1) \rceil$. The result in (16) is a consequence of standard facts about order statistics and you'll prove it on the homework. How do we interpret it? Note, the only thing random in this probability the test point (X_{n+1}, Y_{n+1}) (everything else has been conditioned on). Therefore, we can think about it as follows: *as we draw random calibration sets*, each one being of size n_2 (containing i.i.d. draws

from a fixed distribution P), the coverage integrated over a single test point is distributed as $\text{Beta}(k_\alpha, n_2 + 1 - k_\alpha)$. This distribution has mean

$$\frac{k_\alpha}{n_2+1} = \frac{\lceil (1-\alpha)(n_2+1) \rceil}{n_2+1},$$

exactly as expected. It has variance

$$\frac{k_\alpha(n_2+1-k_\alpha)}{(n_2+1)^2(n_2+2)} \approx \frac{\alpha(1-\alpha)}{n_2+2}.$$

Thus when n_2 is small, this distribution has considerable variability, and for any given calibration set in hand, we might see the test coverage looking far from $1-\alpha$. To give you a more precise sense, Figure 4 plots the density of this beta distribution for $\alpha = 0.1$ and a few values of n_2.

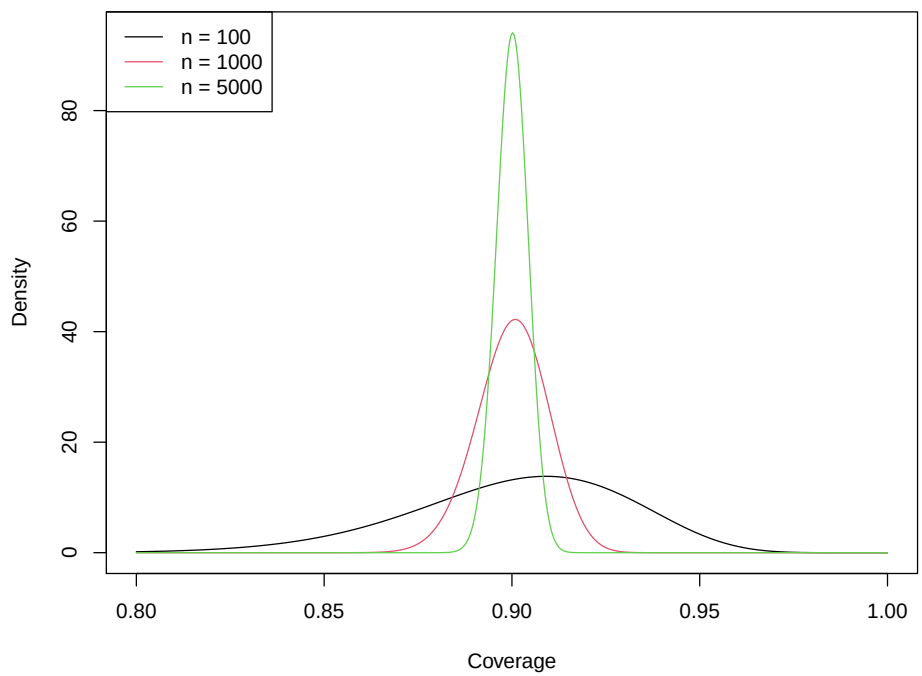

Figure 4: *Density of the beta distribution in* (16) *that describes the calibration set conditional coverage of split conformal prediction, for alpha* $= 0.1$ *and a few values of* n_2.

How about instead conditioning on X_{n+1}? This kind of coverage, which we will call *X-conditional coverage*, would be highly desirable: it would say that

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(x) \,\Big|\, (X_i, Y_i),\, i \in D_1,\, X_{n+1} = x\Big) \geq 1-\alpha, \quad \text{for all } x \in \mathcal{X},$$

which means that we would cover the response at test feature value x. Alas, this is asking for too much, and in a sense that we will make precise later, in Section 3.3, this is effectively impossible to do without making assumptions about the distribution P governing the data.

3 Full conformal prediction

Is there some way to get guaranteed coverage without splitting the data? Enter *full conformal prediction* (often just called conformal prediction). This method is generally much more expensive and much more complicated than its split counterpart, but it is nonetheless a beautiful and important idea—and in some cases, it can indeed be computed efficiently.

In full conformal prediction, we still abide by the second key idea described previously, in which we construct residuals in a way that treats all data symmetrically. We just do it in a more subtle way. Fix any $x \in \mathcal{X}$, and suppose that we want to figure out whether any given response value $y \in \mathbb{R}$ should be in our prediction set $\hat{C}_n(x)$. We call y in the *trial* or *query* value. Now we do something unlike anything we have seen thus far: we train our prediction algorithm on $(X_1, Y_1), \ldots, (X_n, Y_n), (x, y)$—note this is an *augmented* training set, with $n+1$ points—to produce a point predictor $\hat{f}_{n,(x,y)}$. We define residuals

$$\begin{aligned} R_i^{(x,y)} &= |Y_i - \hat{f}_{n,(x,y)}(X_i)|, \quad i = 1, \ldots, n, \\ R_{n+1}^{(x,y)} &= |y - \hat{f}_{n,(x,y)}(x)|. \end{aligned}$$

Then we define a conformal set

$$\hat{C}_n(x) = \Big\{ y : R_{n+1}^{(x,y)} \leq \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } R_1^{(x,y)}, \ldots, R_n^{(x,y)} \Big\}. \tag{17}$$

The guarantee we get is that

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(X_{n+1}) \Big) \in \Big[1-\alpha,\, 1-\alpha+\frac{1}{n+1} \Big), \tag{18}$$

where the lower bound on coverage always holds, and the upper bound holds under the assumption that the residuals are almost surely distinct once we plug in for the random test point $(x, y) = (X_{n+1}, Y_{n+1})$. Why? After we plug in for the random test point, and abbreviate

$$R_i = R_i^{(X_{n+1}, Y_{n+1})}, \quad i = 1, \ldots, n+1,$$

we can see that *these residuals are all exchangeable*. (To be precise, this is only true if the algorithm that we use to fit the point predictor $\hat{f}_{n,(x,y)}$ is a symmetric function of the training data that it takes as input, i.e., does not use knowledge of the order in which the training points were passed.) Therefore

$$Y_{n+1} \in \hat{C}_n(X_{n+1}) \iff R_{n+1} \leq \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } R_i,\, i = 1, \ldots, n$$

occurs (by our previous arguments) with probability at least $1-\alpha$, and at most $1-\alpha+1/(n+1)$ if the residuals are almost surely distinct.

All of the extensions mentioned in the split conformal section, after defining the basic method based on residual scores, carry over to full conformal. We summarize these below.

- Any negatively-oriented and suitably symmetric score function can be used in place of the absolute residual score and the guarantee is unchanged. That is, define

  $$\begin{aligned} R_i^{(x,y)} &= V\Big((X_i, Y_i);\, (X_1, Y_1), \ldots, (X_n, Y_n), (x, y) \Big) \quad i = 1, \ldots, n, \\ R_{n+1}^{(x,y)} &= V\Big((x, y);\, (X_1, Y_1), \ldots, (X_n, Y_n), (x, y) \Big) \end{aligned}$$

  for any function V that is symmetric in its last $n+1$ arguments. This function can, for example, train a point predictor on the last $n+1$ arguments—as long as it treats them symmetrically—and then use it to return some score for the first argument. Then the conformal prediction set in (17) still has the same guarantee in (18), by the same exchangeability arguments.

- We can rewrite the conformal set (17) in equivalent quantile and CDF forms:

  $$\hat{C}_n(x) = \bigg\{ y : R_{n+1}^{(x,y)} \leq \text{Quantile}\bigg(\frac{\lceil (1-\alpha)(n+1) \rceil}{n};\, \frac{1}{n} \sum_{i=1}^n \delta_{R_i^{(x,y)}} \bigg) \bigg\} \tag{19}$$

  $$= \bigg\{ y : \frac{1}{n} \sum_{i=1}^n 1\Big\{ R_i^{(x,y)} \leq R_{n+1}^{(x,y)} \Big\} \leq \frac{\lceil (1-\alpha)(n+1) \rceil}{n} \bigg\}. \tag{20}$$

- We can always inject auxiliary randomness in order to obtain coverage exactly $1-\alpha$ in (18).

3.1 Remarks

The remarks discussed for split conformal previously also carry over more or less to full conformal prediction. To summarize briefly: the full conformal band is protected against overfitting, because now computation of $\hat{f}_{n,(x,y)}$ involves the query point (x, y); the band produced by full conformal under the residual score is often roughly (though not exactly) constant-width, which is not generally a good property, but can be addressed by changing the score function (more later); and lastly, in general, the better the prediction algorithm, the tighter the band will be overall.

Next we give two further remarks. The first one is on computation: full conformal is in general *extremely computationally expensive*: for every x at which we want to compute the prediction set $\hat{C}_n(x)$ in (17), we need to refit the point predictor $\hat{f}_{n,(x,y)}$ (kernel regression, random forest, neural net, etc.) at in principle every $y \in \mathbb{R}$ in order to compute and compare the residuals $R_i^{(x,y)}$, $i = 1, \ldots, n+1$. This would actually be infinitely expensive, but in practice of course we would do it over a finite grid of y values, which could still be tremendously expensive. Relatively speaking, split conformal is computationally trivial: it is typically dominated by the cost of fitting the point predictor $\hat{f}_{n_1}$ once. Due to its extreme computational cost, full conformal prediction is rarely used in practice, except for small problem sizes, or with special prediction algorithms that have something like a "shortcut" formula for refitting the point predictor. Also, it is worth mentioning that in between split and full conformal prediction are methods that look like cross-validation, and cycle through using different parts of the data for training and. See Barber et al. (2021) for details.

The second remark is about interpreting the conformal set via p-values. Observe that we can rewrite (20) once more as

$$\hat{C}_n(x) = \bigg\{ y : \underbrace{\frac{1}{n} \sum_{i=1}^{n} 1\Big\{ R_i^{(x,y)} > R_{n+1}^{(x,y)} \Big\}}_{p(y)} \geq \frac{\lfloor \alpha(n+1) \rfloor}{n} \bigg\}.$$

Informally, we can interpret the fraction of residuals larger than the test residual, which we denote by $p(y)$, as a p-value for the null hypothesis $H_0 : Y_{n+1} = y$. Thus we can think of conformal prediction as using y as a pivotal value, and *keeping all of values y for which we do not reject* the null hypothesis, which it does by comparing $p(y)$ to an adjusted significance level of $\lfloor \alpha(n+1) \rfloor / n$.

3.2 Example

Figure 5 gives an example of full conformal in action. The underlying data is the same as in Figure 3, but for the prediction algorithm we now use a smoothing spline with 15 degrees of freedom (this allows the influence of the query point (x, y) on the spline fit to be more visible). We demonstrate how to calculate the prediction set at a single value $x = 4.75$, marked in blue. Figure 5 is actually an animation, but it might not render as one in many PDF viewers; if you can, try using Adobe Acrobat Reader to see the animation. We run the query response value y over a grid, and for each y, compute the smoothing spline fit using $n+1$ points, the original data set and (x, y). Look carefully and you'll see the gray smoothing spline curve get pulled upwards as y moves upwards. For each value of y, we record the fraction $p(y)$ of residuals larger than the test residual, and in the end, the 90% prediction set keeps y for which $p(y) \geq \lfloor \alpha(n+1) \rfloor / n \approx 0.1$, visualized by thresholding the p-value histogram drawn along the right axis.

3.3 Impossibility of X-conditional coverage

Returning to the issue of X-conditional coverage raised previously, here we drop some disappointing news: conformal prediction methods do not achieve this in general. The story is actually more grim: no method does, in a meaningful way, in the distribution-free setting. If $\hat{C}_n$ is any prediction band such that

$$\mathbb{P}\Big(Y_{n+1} \in \hat{C}_n(x) \,\Big|\, X_{n+1} = x \Big) \geq 1 - \alpha, \quad \text{for any } P, \text{ and } P_X\text{-almost every } x,$$

then Lei and Wasserman (2014) show that

$$\mathbb{P}\bigg(\lim_{\delta \to 0} \sup_{x \in B_\delta(x_0)} \mu(\hat{C}_n(x)) = \infty \bigg) = 1, \quad \text{for any } P, \text{ and any non-atom point } x_0 \text{ of } P_X.$$

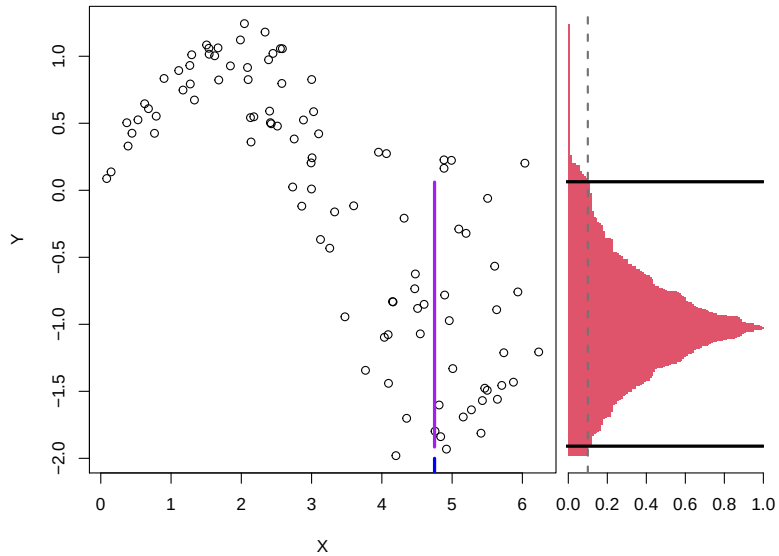

Figure 5: *Example of full conformal prediction, where the prediction algorithm is a smoothing spline with 15 degrees of freedom. (This figure is actually an animation, but it might not render as one in most PDF viewers; if you can, try using Adobe Acrobat Reader to see the animation.)*

As before, μ denotes Lebesgue measure and P_X is the feature distribution associated with P. A non-atom point x_0 of P_X is one for which $P_X(B_\delta(x_0)) \to 0$ as $\delta \to 0$, where $B_\delta(x_0)$ is the ℓ_2 ball of radius δ centered at x_0. Thus, in effect, the above says that any prediction band which claims to cover at almost every x, for every joint distribution P, must be infinite in size at any point x_0 at which we do not have a positive probability of seeing duplicate observations. You'll explore more of the details on your homework.

4 Improving local adaptivity

Even though X-conditional coverage is effectively impossible in the distribution-free setting, in the sense made precise previously, this *does not* mean that different methods cannot have widely different behaviors in practice, when it comes to their ability to deliver approximate X-conditional coverage. We will broaden our terminology and perspective and use *local adaptivity* of a prediction band as a term that refers to its ability to shrink the band at values of x at which prediction is easy, and inflate it at values of x at which prediction is hard. This is admittedly somewhat vague, but because X-conditional coverage isn't really our precise goal, local adaptivity is a better notion to keep in mind.

There are different methods for obtaining better local adaptivity. The ones covered in this lecture all have to do with simply changing the conformity score. Below we cover two options in regression.

4.1 Studentized residuals

A simple variant on the residual score is what we call a *studentized residual*. We describe the idea for split conformal prediction (the full conformal extension is similar). On D_1, we fit both a point predictor $\hat{f}_{n_1}$ and a "spread predictor" $\hat{\sigma}_{n_1}$, which is designed to predict (say) the standard deviation of $|Y - \hat{f}_{n_1}(X)|$ at $X = x$. Then on D_2, we compute "studentized" or normalized residuals

$$R_i = \frac{|Y_i - \hat{f}_{n_1}(X_i)|}{\hat{\sigma}_{n_1}(X_i)}, \quad i \in D_2,$$

and as before $\hat{q}_{n_2} = \lceil (1-\alpha)(n_2+1) \rceil$ smallest of R_i, $i \in D_2$. The conformal set is now

$$\hat{C}_n(x) = \Big[\hat{f}_{n_1}(x) - \hat{\sigma}_{n_1}(x)\hat{q}_{n_2},\, \hat{f}_{n_1}(x) + \hat{\sigma}_{n_1}(x)\hat{q}_{n_2}\Big], \tag{21}$$

whose width we can see adapts according to $\hat{\sigma}_{n_1}$. The guarantee is just as before, in (12). Figure 6 gives an example.

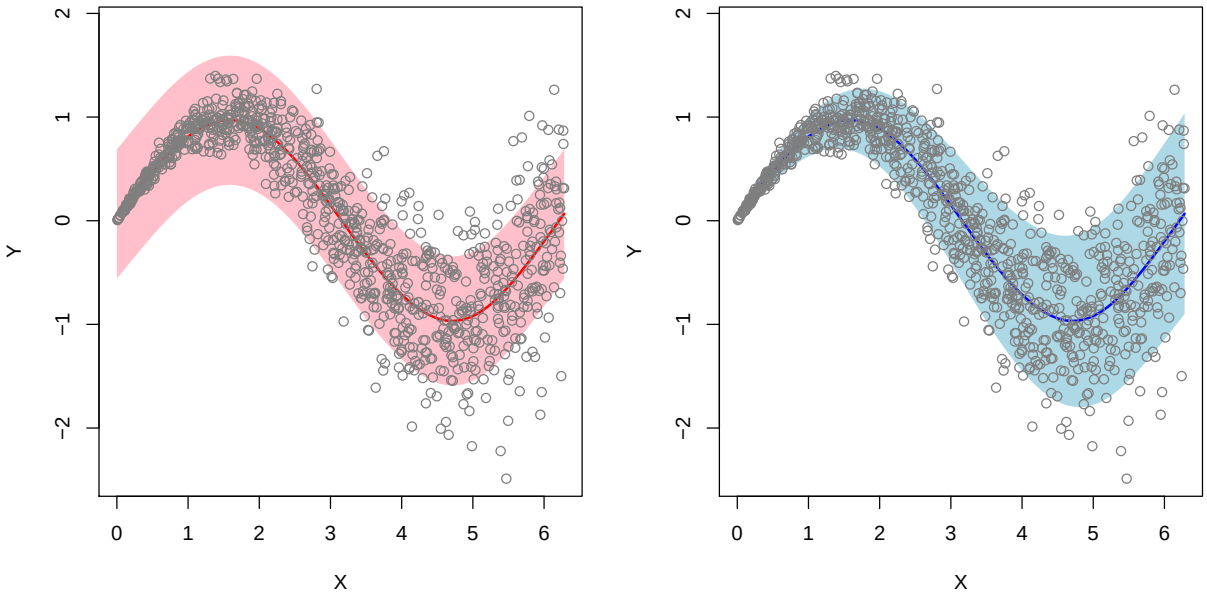

Figure 6: *Examples of split conformal prediction with the usual residual score (left panel) and the studentized residual score (right panel). The data comes from the same generative model as in Figures 3 and 5. We can see that the studentized residual adapts to the local hardness of prediction (and delivers something closer to conditional coverage). Credit: Lei et al. (2018).*

4.2 Quantile regression

There are two issues that can make studentized residuals fall short in practice.

1. If $\hat{f}_{n_1}$ is complex, then little information is left on the proper training set in order to fit $\hat{\sigma}_{n_1}$ (because the proper training residuals will be close to zero). This can be addressed with further data splitting, but that comes at a cost of statistical efficiency.

2. More broadly, it does not need to be true that the variance of $Y|X = x$ and the level $1 - \alpha$ quantile of this distribution are always tied together; in some problem settings they can even have opposing behaviors. Note that the former is being targeted by studentized residuals, but the latter is what should really be targeting in prediction bands.

Romano et al. (2019) show that both of these can be addressed by changing our perspective on the point predictor itself: why not have it predict the level $1 - \alpha$ quantile of the response at $X = x$ directly? (As opposed to predicting the mean of the response at $X = x$, which is what generic regression methods do.)

Their proposal, called *conformalized quantile regression* (CQR), works as follows, in the split setting (the full version is similar). We first fit two point predictors, denoted $\hat{f}_{n_1}^{\alpha/2}$ and $\hat{f}_{n_1}^{1-\alpha/2}$, on the proper training data (X_i, Y_i), $i \in D_1$. Here each $\hat{f}_{n_1}^{\tau}(x)$ is estimates the level τ quantile of $Y|X = x$. This can be obtained from a variety of quantile regression methods, which often only require a change of the loss function from a generic regression method. Then we form calibration set scores

$$R_i = \max\Big\{\hat{f}_{n_1}^{\alpha/2}(X_i) - Y_i,\, Y_i - \hat{f}_{n_1}^{1-\alpha/2}(X_i)\Big\}, \quad i \in D_2,$$

and as before $\hat{q}_{n_2} = \lceil (1-\alpha)(n+1) \rceil$ smallest of R_i, $i \in D_2$. The CQR set is now

$$\hat{C}_n(x) = \Big[\hat{f}_{n_1}^{\alpha/2}(x) - \hat{q}_{n_2},\, \hat{f}_{n_1}^{1-\alpha/2}(x) + \hat{q}_{n_2}\Big]. \tag{22}$$

which enjoys the same guarantee as in (12).

In the example from Figure 6, the data set is large enough and the point predictor stable enough that CQR (not shown) provides little gain over studentized residuals. However, in other examples, it can provide a clear gain. See Romano et al. (2019). Still, studentized residuals (or variants thereof) remain in fairly common use because we can use "out-of-the-box" regression methods to fit each of $\hat{f}_{n_1}, \hat{\sigma}_{n_1}$ in sequence.

5 Conformal classification

In this last section, we briefly cover conformal prediction for classification problems. The story is much the same, but we need different conformity score functions, since residual or quantile regression scores are not generally appropriate in classification. Below we cover a standard choice, based on predicted probabilities. Then we cover a choice that is designed to have better local adaptivity, based on cumulative probabilities. Throughout, we take $\mathcal{Y} = \{1, \dots, K\}$.

5.1 Likelihood scores

We describe the idea for split conformal formulation (full conformal being similar). We first fit a probabilistic classifier $\hat{f}_{n_1}$ to the proper training data (X_i, Y_i), $i \in D_1$. That is, to be clear,

$$\hat{f}_{n_1}(x; k) \text{ estimates } \mathbb{P}(Y = k \,|\, X = x), \text{ for each } k = 1, \dots, K.$$

We then form calibration set scores,

$$R_i = \hat{f}_{n_1}(X_i; Y_i), \quad \in D_2.$$

In words, each R_i is the probability or likelihood assigned to the correct class (on the unseen observation (X_i, Y_i) from the calibration set). Thus note that this is a positively-oriented score. We let

$$\hat{q}_{n_2} = \lfloor \alpha(n_2 + 1) \rfloor \text{ smallest of } R_i,\ i \in D_2,$$

and define the conformal set

$$\hat{C}_n(x) = \Big\{ k : \hat{f}_{n_1}(x; k) \geq \hat{q}_{n_2} \Big\}.$$

This has precisely the same guarantee as in (12).

5.2 Cumulative likelihood

To make the conformal prediction sets more adaptive (still in the context of having fit a probabilistic classifier $\hat{f}_{n_1}$ on D_1), Romano et al. (2020) propose a conformity score based on cumulative likelihood, defined as follows. For each $i \in D_2$, let π_i be the permutation of $1, \dots, K$ that sorts the predicted probabilities $\hat{f}_{n_1}(X_i; k)$, $k = 1, \dots, K$ in decreasing order, so that

$$\hat{f}_{n_1}(X_i; \pi_i(1)) \geq \hat{f}_{n_1}(X_i; \pi_i(2)) \geq \dots \geq \hat{f}_{n_1}(X_i; \pi_i(K)).$$

The the conformity scores are

$$R_i = \sum_{j=1}^{k_i} \hat{f}_{n_1}(X_i; \pi_i(j)), \quad \text{where } \pi_i(k_i) = Y_i, \text{ for each } i \in D_2.$$

In words, each R_i is the cumulative probability of all classes considered "at least as likely" as the true one, by our probabilistic classifier. Note that this is negatively-oriented (if the true class is assigned a very low probability, then the cumulative probability of all classes "at least as likely" as it will be very high). Hence we let

$$\hat{q}_{n_2} = \lceil (1 - \alpha)(n_2 + 1) \rceil \text{ smallest of } R_i,\ i \in D_2,$$

and define the conformal set

$$\hat{C}_n(x) = \{\pi_x(1), \ldots, \pi_x(k_x)\}, \quad \text{where } k_x = \min\left\{k : \sum_{j=1}^{k} \hat{f}_{n_1}(x; \pi_x(j)) \leq \hat{q}_{n_2}\right\}.$$

This has precisely the same guarantee as in (12).

Angelopoulos et al. (2021) refer to this method as *adaptive prediction sets* (APS) and define a regularized version called RAPS that often delivers much smaller sets in practice. Figure 7 gives a few examples of RAPS from their paper.

Figure 7: *Examples of RAPS, which is conformal prediction with a regularized version of the cumulative likelihood score. The true label in each case is "fox squirrel", and we can see that the prediction sets adapt appropriately in size to the hardness of the classification task. Credit: Angelopoulos et al. (2021).*

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